

FED-LAMP: A FEDERATED APPROACH TO RELABEL NEUROIMAGING DATA

*Raviteja Girijala*¹ *Sunitha Basodi*^{1*} *Theo G.M. van Erp*^{2,3} *Bradley Baker*¹
*Jessica A Turner*⁴ *Vince D. Calhoun*¹

¹ TReNDS Center, Georgia State University, Georgia Institute of Technology, Emory University, Atlanta, USA

² Department of Psychiatry and Human Behavior, School of Medicine, University of California, Irvine, CA 92617, USA

³ Center for the Neurobiology of Learning and Memory, University of California, Irvine, CA 92617, USA

⁴ Department of Psychiatry and Behavioral Health, Wexner Medical Center, The Ohio State University, Columbus, OH, USA

ABSTRACT

Label-noise filtering-based dimensional prediction (LAMP) is a robust method for handling label noise in multi-site neuroimaging. It identifies typical subjects whose imaging patterns agree with their clinical labels, constructs robust class prototypes, and uses these prototypes to score and refine noisy labels. This yields cleaner cohorts for downstream analyses of functional network connectivity (FNC) derived from resting-state functional Magnetic Resonance Imaging (fMRI). However, the need to aggregate sensitive raw data limits LAMP’s applicability across institutions. Therefore, we introduce Federated LAMP, a privacy-preserving federated variant that executes typical-subject selection, site-wise confidence calibration, and prototype updates locally while sharing only aggregated statistics or model parameters without transmitting the original data. Federated LAMP is designed for a privacy-preserving aggregator that aligns global prototypes without moving raw data. The workflow is implementation-agnostic and fits standard federated infrastructures, and provides a scalable, privacy-compliant framework for neuroimaging analyses.

Index Terms— Federated learning, Functional Network Connectivity (FNC), adaptive relabeling, Schizophrenia

1. INTRODUCTION

Large multi-site neuroimaging cohorts are essential for detecting reproducible brain-behavior effects and testing shared hypotheses, but three issues that repeatedly limit joint analyses, namely: (i) site effects from acquisition and scanner differences, which inflate variance and bias; (ii) missing values introduced during pre-processing, which complicate group-level statistics; and (iii) governance constraints (Data Use Agreements, de-identification) that slow or prevent data aggregation [1][2]. We propose a federated variant of LAMP that couples privacy-preserving coordination with label-quality control and dimensional scoring. Each participating

site locally identifies “typical” subjects—those whose imaging features align with their label—handles missingness with lightweight linear models or masking, and computes compact statistics. Only these summaries are shared to form global centroids and train a dimensional predictor; raw images never leave the institution. This design is compatible with standard harmonization, but shifts learning toward the most reliable signal while respecting local policies. In practice, federated LAMP tightens within-group compactness, enlarges inter-group separation, and improves cross-site generalization, offering a practical path to scalable and privacy-preserving discovery in neuroimaging.

2. PRELIMINARIES

In this section, we provide a brief description of a filtering-based label-noise prediction algorithm and federated learning.

2.1. Federated Learning Setup

Federated learning enables multi-site collaboration without sharing raw imaging data: each site performs local computations such as gradients, feature statistics, and model parameters, and transmits only privacy-preserving aggregates or parameters to a coordinating service that securely combines them and returns an updated global state [3].

2.2. LAMP (label-noise filtering-based dimensional prediction)

Our method provides a federated algorithm for the existing centralized label filtering approach, LAMP [4]. Below are some of the key steps performed in LAMP that we will be federating in our methods section. For a given labeled neuroimaging data, the authors in [4] perform identification of subjects that have reliable labels assigned (named typical subjects) for each cohort, cross-check label reliability across cohorts, and identify a threshold based on this analysis to relabel all the data.

*Corresponding author: sbasodi1@gsu.edu

2.2.1. Identification of typical subjects

In each cohort, complete random forest(CRF) is used to identify the subjects whose labels are reliable and do not vary across various CRF's. After identifying these subjects, named typical subjects, feature selection algorithms are used to reduce the data dimension. In [4], 2-sample t-tests are used for feature selection. Typical subjects with reduced features are then used for intra-site label analysis.

2.2.2. Inter-site label validation

The typical subjects are grouped by label and are averaged to form label-based centroids for a cohort. This indicates how closely subjects with the same label cluster together after re-labeling [5]. If compactness improves, subjects of a label tend to look more similar to one another and less similar to subjects of the other label. In practice, we check that the spread of each label around its own centroid. Now each cohort compares its subject data with the centroids of other cohorts, which is then used in the next step for identifying label thresholds.

2.2.3. Thresholding and re-labeling the data

After computing distance scores from other cohort centroids, all the data is sorted to identify the boundaries to perform re-labeling. Threshold (t) is computed by taking the average of the minimum value in the positive label and the maximum value in the negative label. Data in all the cohorts is then re-labeled as below:

$$\text{Relabel}(\text{Subject}_i; t) = \begin{cases} \text{HC}, & \text{if } \text{score}_i < -t, \\ \text{SZ}, & \text{if } \text{score}_i > t, \\ \text{excluded}, & \text{otherwise.} \end{cases} \quad (1)$$

3. FED-LAMP: A FEDERATED REFORMULATION TO RELABEL NEUROIMAGING DATA

Here, we describe how the LAMP workflow is redesigned in a federated manner so that each cohort, represented as a separate site, performs all subject-level operations locally while only sharing model summaries. The goal is to develop a federated algorithm keeping the original principles of label-noise filtering and dimensional scoring intact (as explained in 2.2) The overall flow is illustrated in Fig. 1. We employ algorithms used in [4] to determine typical subjects, compute inter-site scores, compute adaptive thresholding, and relabel the data.

3.1 Local Identification of Typical Subjects

Each site begins by filtering the subjects using complete random forest models. This step measures the stability of the original label across the labels predicted by these models. Subjects with more consistent label trajectories are treated as typical, which follows the same motivation as earlier work

on dependable label discovery [4]. After determining typical subjects, each site performs a simple feature selection using two-sample t-tests in healthy and patient groups. This produces a reduced feature set that better reflects the group differences. Using only the typical subjects with their selected features, each site computes an average feature vector for both the healthy and the patient group. These local summaries that represent the site-specific information are then shared with the aggregator (remote) site, and therefore, no subject-level data will be shared.

3.2 Forming Global Models

The remote aggregator receives two group averages sent by each of the participating sites. These are aggregated across sites to form a map of centroids for healthy and patient groups. The remote aggregator then sends these global references back to every site. Because only aggregated centroids are transmitted, this stage preserves privacy while still allowing every site to use the same shared global information.

3.3 Computing Dimensional Scores at Each Site

After receiving the global models, each site computes a dimensional score for every subject. This score quantifies how similar a subject is to the patient versus a healthy reference pattern. First, each subject measures its distance from the healthy and patient centroids sent by the aggregator. Then each subject computes its per-source score using, and finally, each site aggregates over all sources using to obtain a single-dimensional score for every subject. This score places the subject along a continuous healthy-patient axis and reduces thousands of imaging features to a single interpretable value.

3.4 Federated Adaptive Thresholding

Dimensional score ranges vary across sites and iterations. In the federated method, each site sends only its dimensional scores to a remote aggregator. The aggregator combines all scores from all sites and computes one global threshold. This global threshold is returned to all sites. Each site then applies the same thresholding rule to determine whether a subject is strongly healthy, strongly patient, or falls in the boundary region. This ensures that relabeling decisions are consistent across all sites while maintaining strict privacy.

3.5 Producing Updated Labels

Once the new labels are assigned, each site updates its typical subjects and recomputes its group averages. These updated summaries are the input for the next round. Because the dimensional score, thresholding, and prototype updates are performed on aggregated information, no original data leaves the institution at any stage. Over iterations, typical subjects become cleaner, intra-group compactness improves, and inter-group differences become more stable. As shown in earlier findings [4], this refinement leads to clearer separation between the healthy and patient groups, both within and across datasets.

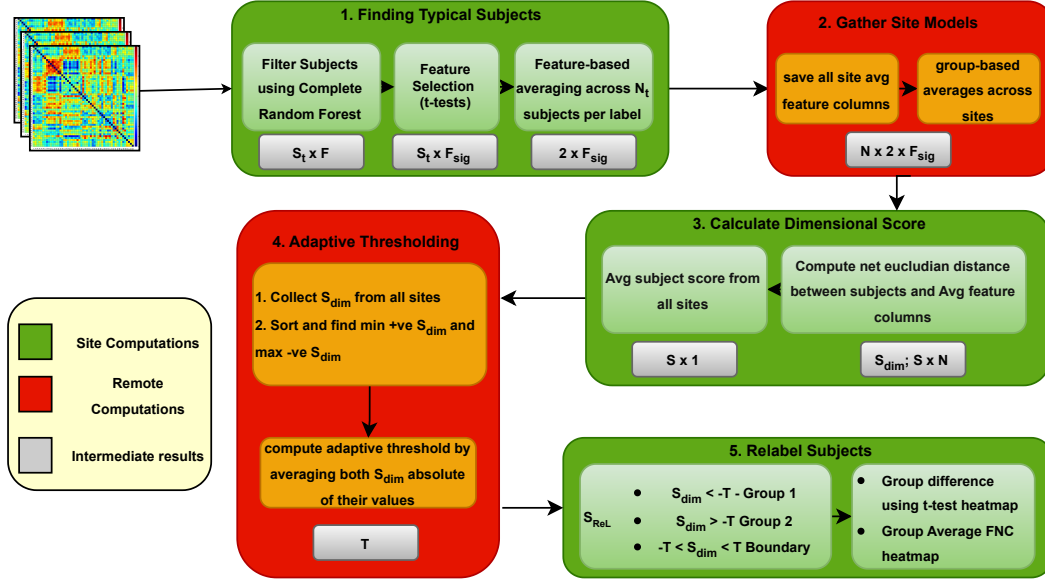


Fig. 1. Overall flow of federated LAMP along with the steps at each iteration.

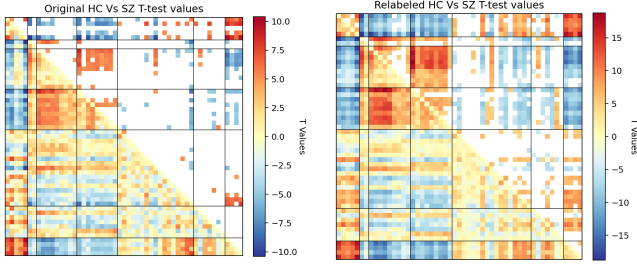


Fig. 2. Figure 1. Original HC vs SZ (left) and Relabeled HC vs SZ (right) two-sample t -test T -value maps. Upper triangle shows edges with $p < 0.01$ after Bonferroni correction; lower triangle shows uncorrected t values.

4. EXPERIMENTS AND RESULTS

4.1. Datasets and Typical Subjects

We evaluated Fed-LAMP using two datasets, COBRE [6] and FBIRN [7], as separate participating sites containing patients with SZ and healthy controls. COBRE consists of 157 subjects (SZ: 68 and HC: 89), and FBIRN consists of 311 subjects (SZ: 151 and HC: 160). For each of these datasets, we first compute the CRF step locally at each site to identify the *typical* subjects shown in Table 1.

Dataset	Total Typical	Typical SZ	Typical HC
FBIRN	191	99	92
COBRE	103	55	48

Table 1. Number of typical subjects identified in each dataset.

4.2. Two-sample t -tests on FNC (Original vs. Relabeled)

We compare group differences before and after relabeling using edge-wise two-sample t -tests on the FNC matrices of re-labeled subjects:

- **Original groups:** HC vs. SZ using the dataset’s provided labels.
- **Relabeled (typical) groups:** ReHC vs. ReSZ using Fed-LAMP labels.

In Fig. 2, the **upper triangle** shows significant values at $p < 0.01$ with Bonferroni correction (more conservative, emphasizes robust effects). The **lower triangle** shows the corresponding uncorrected t -values (helps visualize the overall pattern). The t -maps summarize where the two groups differ. This shows that the relabeled data is reliable as the map gets clearer, has consistent patterns after relabeling, such as stronger magnitudes, few scattered isolated edges, and greater cross-dataset consistency. In short, sharper t -maps after relabeling mean the groups are cleaner and more separable.

The comparison between the centralized and federated relabeling results shows that both methods produce almost identical functional network connectivity patterns for the typical subjects, as shown in Fig. 3. The first plot (Fig. 3(a)) com-

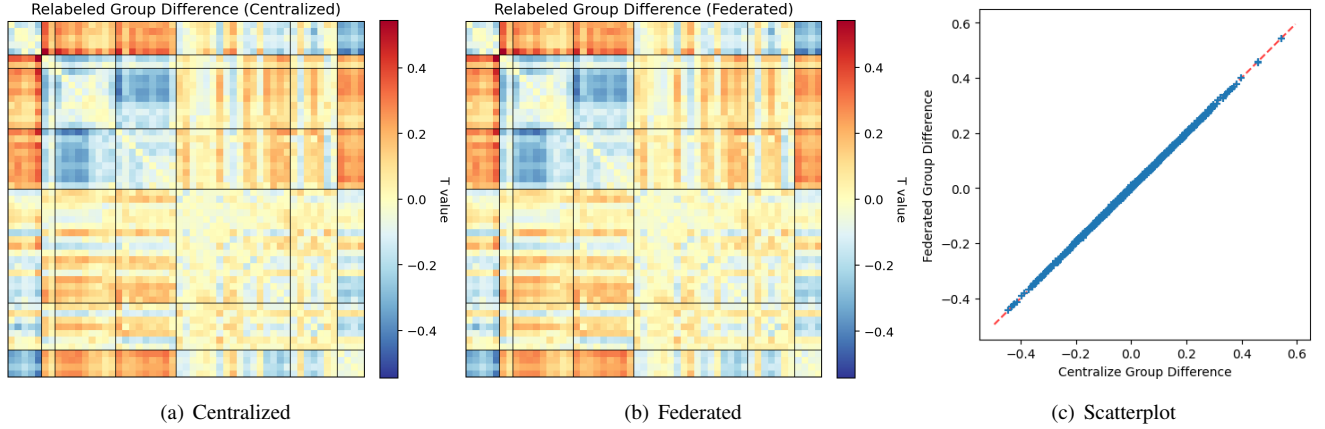


Fig. 3. Comparison of Centralized and Federated group difference maps for relabeled data along with their differences

compares the relabeled healthy control group obtained through the centralized LAMP pipeline with the group obtained using the federated workflow. The second plot (Fig. 3(b)) shows the same comparison for the relabeled patient group in the fed-LAMP. Fig. 3(c) shows the scatter plot of these FNC values with centralized FNC data on the X-axis and the corresponding federated FNC data on the Y-axis. The points lie along the identity line, which indicates that the average FNC values produced by both approaches match one another without noticeable deviation. Together, these results show that the federated implementation preserves the original behavior of the centralized method.

5. CONCLUSION

In this work, we present a federated algorithm to relabel neuroimaging data using LAMP, which does not require any actual data sharing to perform a combined analysis. Each site selects typical subjects, builds group centroids, and computes dimensional scores while sharing only summaries without sending actual data. The federated model delivers the same results compared to its centralized model and reduces label noise, and yields stronger group differences than the original setup.

6. ACKNOWLEDGEMENTS

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